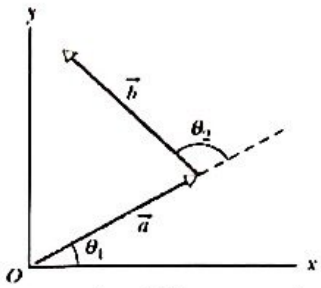
MNS Department
QUIZ - 1
Semester: Spring-2024; Section - 22
Course ID: PHY 111; Course Title: Principles of Physics I
Total Marks: 15

Name: ***ID:***

1. The two vectors in figure have equal magnitudes of 15.0 m and the angles are $\theta_1 = 20.0°$ and $\theta_2 = 115.0°$.



   (a) (2 marks) Find the vectors in unit vector notation.
   (b) (3 marks) Find the 3rd vector $\vec{d}$ such that $(2\vec{b} \times \vec{a}) + (\vec{d} + 3\vec{b} - \vec{a}) = 0$.
   (c) (3 marks) Find the angle between $\vec{d}$ and $\vec{b}$.

2. (4 marks) The position of a car is given by $x = 12t - 3t^2 + 4$. Find the (i) velocity of that car at 3 sec and (ii) average acceleration between time $t = 2s$ to $t = 5s$. [x is in meters and t is in seconds]
3. (2 marks) Vector $\vec{d_1}$ is in the negative direction of a y axis, and vector $\vec{d_2}$ is in the positive direction of a z axis. What is the direction of the vector resulting from (i) $\vec{d_1} \times \vec{d_2}$ and (ii) $\vec{d_2} \times \vec{d_1}$.

$-\hat{i}$ $\boxed{-x}$ $\hat{i}$ $\boxed{+x}$

4. (1 mark) Here are four descriptions of the position of a puck as it moves in a straight line:
   (1) $x = -4t^{-2} + 2t$ (2) $x = -4t - 8$ (3) $x = 3t - 2$ (4) $x = 3/t + 5$
   Which puck (pucks) is (are) moving at constant velocity?

(1)
(a) $\vec{a} = 15\cos 20°\,\hat{i} + 15\sin 20°\,\hat{j} = 14.1\hat{i} + 5.13\hat{j}$

$\vec{b} = 15\cos 135°\,\hat{i} + 15\sin 135°\,\hat{j} = -10.61\hat{i} + 10.61\hat{j}$

(b) $(2\vec{b} \times \vec{a}) + (\vec{d} + 3\vec{b} - \vec{a}) = 0$

$\Rightarrow \vec{d} = \vec{a} - 3\vec{b} - (2\vec{b} \times \vec{a})$

$= 14.1\hat{i} + 5.13\hat{j} + 31.83\hat{i} - 31.83\hat{j} + 408.06\hat{k}$

$\therefore \vec{d} = 45.93\hat{i} - 26.7\hat{j} + 408.06\hat{k}$

$2\vec{b} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -21.22 & 21.22 & 0 \\ 14.1 & 5.13 & 0 \end{vmatrix}$

$= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(-108.86 - 299.2)$

$= -408.06\hat{k}$

(c) $\theta = \cos^{-1}\left(\frac{\vec{d} \cdot \vec{b}}{d\,b}\right)$

$= \cos^{-1} \frac{-770.61}{15 \times 411.5}$

$\therefore \theta = 97.17°$

$\vec{d} \cdot \vec{b} = (45.93\hat{i} - 26.7\hat{j} + 408.06\hat{k}) \cdot (-10.61\hat{i} + 10.61\hat{j})$

$= -487.32 - 283.29$

$= -770.61$

$b = 15$

$d = \sqrt{(45.93)^2 + (-26.7)^2 + (408.06)^2}$

$= 411.5$

(2) $x = 12t - 3t^2 + 4$

$V = 12 - 6t$

(i) $V_3 = 12 - 6 \times 3 = \boxed{-6\,ms^{-1}}$

(ii) $a_{avg} = \frac{\Delta V}{\Delta t} = \frac{V_5 - V_2}{5 - 2} = \frac{-18 - 0}{3} = \boxed{-6\,ms^{-2}}$

$V_2 = 12 - 6 \times 2 = 0$

$V_5 = 12 - 6 \times 5 = -18$